



NAME

CT GROUP

20S

CHEMISTRY

Paper 1 Multiple Choice

9729/01

1 October 2020

40 minutes

Additional Materials: Multiple Choice Answer Sheet
Data Booklet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, glue or correction fluid.

Complete the information on the Answer Sheet as shown below.

1. Enter your **NAME** (as in NRIC). _____

2. Enter the **PAPER NUMBER**. _____

3. Enter your **CT GROUP**. _____

4. Enter your **NRIC NUMBER** or
FIN Number

5. Now **SHADE** the corresponding
circles in the grid for
EACH DIGIT or **LETTER**

USE PENCIL ONLY							
FOR ALL ENTRIES ON THIS SHEET							
0	1	2	3	4	5	6	7
☐	☐	☐	☐	☐	☐	☐	☐

NRIC / FIN											
(S)	(0)	(0)	(0)	(0)	(0)	(0)	(0)	(A)	(K)	(U)	
(F)	(1)	(1)	(1)	(1)	(1)	(1)	(1)	(B)	(L)	(V)	
(G)	(2)	(2)	(2)	(2)	(2)	(2)	(2)	(C)	(M)	(W)	
(T)	(3)	(3)	(3)	(3)	(3)	(3)	(3)	(D)	(N)	(X)	

There are **twenty** questions on this paper. Answer **all** questions. For each question, there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

The use of an approved scientific calculator is expected, where appropriate.

- 1 ^{235}U , uranium-235, is a radioactive element that can release 'fast neutrons'.

How many neutrons are present in a 2.35 g sample of ^{235}U ?
[L = Avogadro constant]

- A 0.92 L B 1.43 L C 1.46 L D 2.35 L

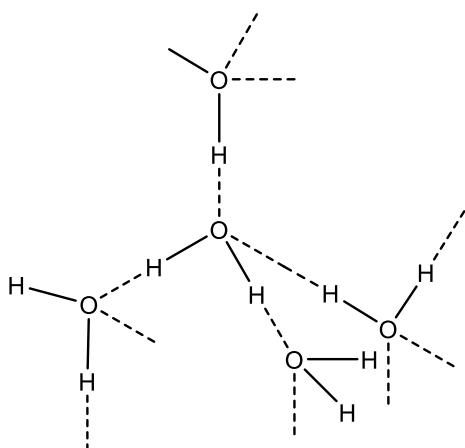
- 2 Three solid substances have the following properties.

	X	Y	Z
Melting point / $^{\circ}\text{C}$	800	110	1085
Conductivity of solid at room temperature	nil	nil	high
Solubility in water	soluble	poorly soluble	insoluble

What is the likely identity of X, Y and Z?

	X	Y	Z
A	NaCl	iodine	Cu
B	NaCl	CCl_4	SiO_2
C	Cu	NH_3	graphite
D	HCl	CCl_4	graphite

- 3 The diagram shows the structure of part of a crystal of ice.



Which statements are correct about the lattice structure of ice?

- The H–O–H bond angle is larger in ice than in water.
- Each H_2O molecule is involved in four hydrogen-bonds.
- When ice melts, O–H covalent bonds are broken.

- A 1 and 2 only
B 2 and 3 only
C 1, 2 and 3
D 1 only

4 Which of the following is **incorrect** about an ideal gas?

- A Its particles are point masses with no molecular volume.
- B Its particles have no attraction or repulsion for each other.
- C There is no conversion of energy during collisions between particles.
- D Changing its temperature has no effect on the kinetic energy of the particles.

5 *Use of the Data Booklet is relevant to this question.*

Sodium percarbonate, $(\text{Na}_2\text{CO}_3)_x \cdot y(\text{H}_2\text{O}_2)$, is an oxidizing agent in some home and laundry cleaning products.

On acidification, 10.0 cm^3 of $0.100 \text{ mol dm}^{-3}$ sodium percarbonate releases 48.0 cm^3 of carbon dioxide at room conditions.

An identical sample, on titration with $0.0500 \text{ mol dm}^{-3} \text{KMnO}_4$, requires 24.0 cm^3 before the first pink colour appears. KMnO_4 reacts with H_2O_2 in the mole ratio 2:5.

What is the ratio y/x ?

- | | | | |
|-----------------|-----------------|-----------------|-----------------|
| A $\frac{1}{3}$ | B $\frac{2}{3}$ | C $\frac{3}{2}$ | D $\frac{6}{2}$ |
|-----------------|-----------------|-----------------|-----------------|

6 A 15 cm^3 sample of propene gas, C_3H_6 , was burnt in 60 cm^3 of oxygen to produce a mixture of carbon dioxide and carbon monoxide. In the process, the total volume of gases contracted by 30 cm^3 and no propene or oxygen remained. All volumes were measured at the same temperature and pressure.

What can be deduced from the information given?

- 1 When the resulting gas mixture was passed through aqueous sodium hydroxide, 30 cm^3 of gas was absorbed.
- 2 At the temperature at which the gas volumes were measured, the H_2O produced was not in the gaseous state.
- 3 Only one third of the propene gas underwent complete combustion.

- A 1 and 2 only
- B 1 and 3 only
- C 1, 2 and 3
- D 2 only

7 Which compound has the largest magnitude of lattice energy?

- | | | | |
|-------|-------|-------|-------|
| A CaS | B CaO | C KBr | D KCl |
|-------|-------|-------|-------|

8 Which statement about the dissolution of ionic solids is correct?

- A As temperature increases, ΔG_{sol} becomes more positive.
- B The standard enthalpy change of solution is always exothermic.
- C The standard enthalpy change of hydration is always exothermic.
- D The greater the magnitude of lattice energy, the more soluble the ionic solid will be

9 Which processes have a positive ΔS ?

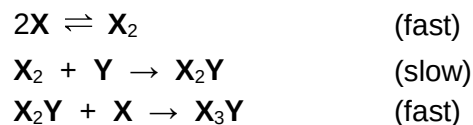
- 1 $\text{H}_2\text{O}(l) \rightarrow \text{H}_2\text{O}(s)$
- 2 $2\text{Fe}^{3+}(g) + 3\text{O}^{2-}(g) \rightarrow \text{Fe}_2\text{O}_3(s)$
- 3 $\text{Ar}(g) \text{ at } 25^\circ\text{C} \rightarrow \text{Ar}(g) \text{ at } 50^\circ\text{C}$

- A 1 and 2 B 2 and 3 C 3 only D 1, 2 and 3

10 Consider the following reaction:



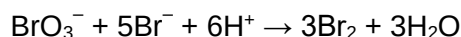
The proposed mechanism for the reaction involves the following steps:



Based on the information, what is the rate equation for this reaction?

- | | | | |
|---|----------------------------------|---|----------------------------------|
| A | Rate = $k[\text{X}]^2[\text{Y}]$ | C | Rate = $k[\text{X}]^2$ |
| B | Rate = $k[\text{X}_2][\text{Y}]$ | D | Rate = $k[\text{X}]^3[\text{Y}]$ |

11 Colourless BrO_3^- reacts with Br^- to form coloured Br_2 according to the equation below:



Which methods could be used to monitor the progress of the reaction?

- 1 Using a colorimeter to measure the colour intensity of the solution.
- 2 Measuring the electrical conductivity of the solution.
- 3 Measuring the pH of the solution.

- A 1 and 2 B 2 and 3 C 3 only D 1, 2 and 3

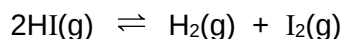
12 Fe^{2+} acts as a homogeneous catalyst in the reaction between I^- and $\text{S}_2\text{O}_8^{2-}$. Which statement is **incorrect** about the effect of Fe^{2+} on this reaction?

- A It increases the value of the rate constant.
- B It does not appear in the rate equation for the reaction.
- C It causes the reaction to proceed via a different mechanism.
- D It provides an alternative pathway of lower activation energy for the reaction to proceed.

13 Which statement is correct for a system in dynamic equilibrium?

- A Equilibrium can be achieved in an open system.
- B The concentration of the reactants and products change.
- C The rates of the forward and backward reactions are non-zero.
- D Adding a catalyst causes the position of equilibrium to change.

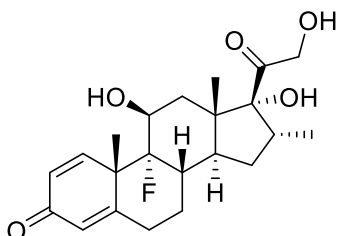
14 The value of the equilibrium constant, K_p , for the following reaction is 4.0 at T °C.



When a sample of HI is decomposed in a sealed tube at T °C and 1.0 atm, what is the equilibrium partial pressure of H_2 in the tube?

- A 0.40 atm B 0.47 atm C 0.57 atm D 0.63 atm

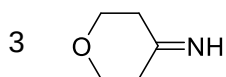
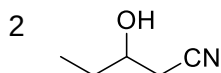
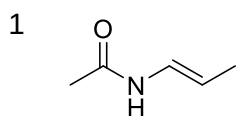
15 Dexamethasone is an anti-inflammatory steroid which also has the ability to suppress a person's immune system. It is being trialed in the US and UK as a possible drug for COVID-19 treatment.



Which functional group is **not** found in Dexamethasone?

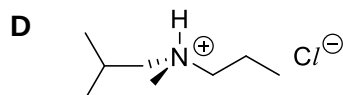
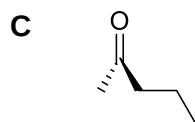
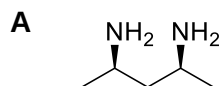
- A Alcohol
- B Alkene
- C Carboxylic acid
- D Halogenoalkane

16 Which molecules are constitutional isomers of each other?



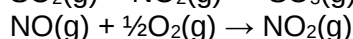
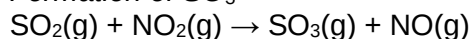
- A 1 and 2
 B 2 and 3
 C All of the above
 D None of the above

17 Which compound rotates plane polarised light?

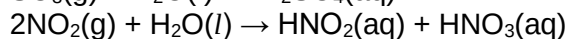
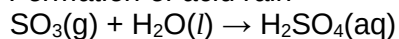


- 18** SO_2 and NO_2 are pollutants formed during combustion in motor vehicles. They then react in the atmosphere to form SO_3 or acid rain.

Formation of SO_3

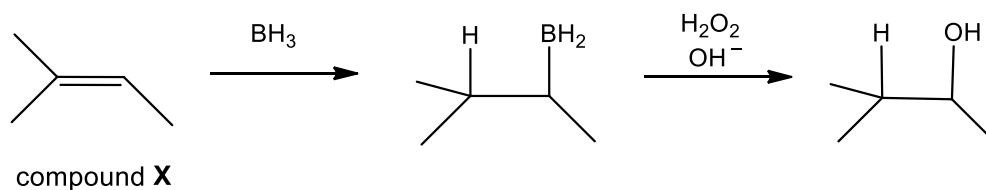


Formation of acid rain



Based on the information given, which statement is **incorrect**?

- A** Acid rain contains a mixture of acids.
- B** The given reactions are all redox reactions.
- C** NO_2 undergoes disproportionation in the atmosphere.
- D** NO_2 acts as a homogeneous catalyst in the formation of SO_3 .
- 19** Hydroboration is a two-step reaction to prepare alcohols from alkenes. Compound **X** undergoes hydroboration as follows.

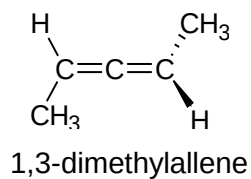


Based on the information given, which statement is correct?

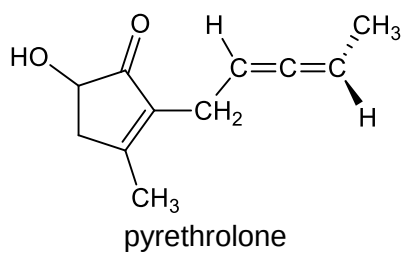
- A** H_2O_2 acts as a reducing agent in the second step.
- B** The type of reaction that occurs in the first step is substitution.
- C** In the first step, the boron atom of BH_3 forms a bond with the more substituted carbon of the $\text{C}=\text{C}$ group.

- D** would be formed if BD_3 was used instead of BH_3 in the first step. ($\text{D} = {}^2_1\text{H}$)

- 20** 1,3-dimethylallene exists as a pair of enantiomers because the carbon atoms at each end of the C=C=C functional group are attached to two non-identical substituents.



Pyrethrolone is the first natural allene that was isolated in 1924.



Which statement about pyrethrolone is correct?

- A** It is a planar molecule.
- B** It has 2^3 stereoisomers.
- C** It is not susceptible to attack by electrophiles.
- D** It contains five sp^2 hybridised carbon atoms.

~ END OF PAPER ~

Answers

Final

1	2	3	4	5	6	7	8	9	10
B	A	A	D	C	A	B	C	C	A

11	12	13	14	15	16	17	18	19	20
D	B	C	A	C	C	D	B	D	D